



UNIVERSITY OF PERADENIYA
DEPARTMENT OF STATISTICS AND COMPUTER SCIENCE
END SEMESTER EXAMINATION - SEMESTER II (2019/2020) – JANUARY 2022
CS 105 – OBJECT-ORIENTED PROGRAMMING PRACTICAL (1 CREDIT)

Time Allowed: **Three** Hours

Answer **ALL** Questions

1)

A Taxi company needs to develop an application to achieve the following tasks.

Task 1: Get the location details where the location distance is less than a given value.

Task 2: Get the Taxi Details of the Taxi's having the highest and second highest number of passengers.

Implement a java program to achieve the above tasks,

- a) Create a method “**searchByDistance()**” to return the location details as stated in **Task 1**.
(**Note: Additional marks will be given for handling exceptions)
- b) Create a method “**getTwoMaxPassengerTaxis()**” to return the Taxi details as stated in **Task 2**.
- c) Implement a driver class and test the two created methods in part a) and b) using the details given below.

Taxi Details:

Taxi ID	Number of Passengers
Taxi 001	3
Taxi 002	4
Taxi 003	6
Taxi 004	2
Taxi 005	5

Location Details:

Location Name	Location Distance
Texas	250.0
Boston	65.0
New York	80.5
San Diego	15.8
Huston	160.0

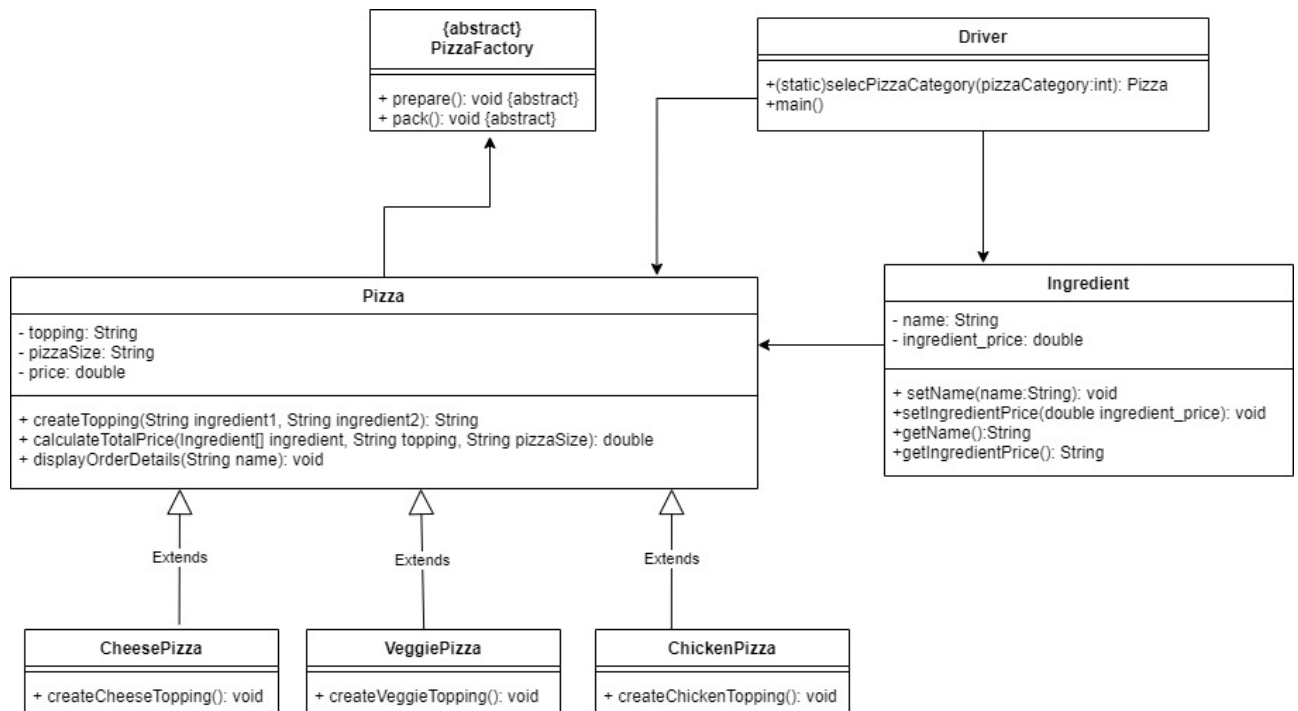
[40 Marks]

[P.T.O]

2)

A Pizza Factory needs to implement an application to manage their daily orders. This application lets users enter the name, the pizza size, pizza category and also the extra ingredients required for their topping of choice. Finally, the application displays the details of the pizza bought by the customer and shows its total price.

a) Implement a java application to carry out the above process using the class diagram below:



PizzaFactory

Contains two abstract methods **prepare()** and **pack()** where it prints the status of the order as “**preparing your order**” and “**order is ready**”.

Ingredients

Has two variables **name** and **ingredient_price** to store the name and price of all the ingredients available in the Pizza Factory.

Pizza

Contains the following methods as given below:

createTopping() - Take two ingredients as parameters and append the topping and return a single string in the following format: **topping,ingredient1,ingredient2**

calculateTotalPrice() - Based on the created topping and pizza size you should calculate the total price of the Pizza.

displayOrderDetails() - Display all the details as given below.

[P.T.O]

CheesePizza, VeggiePizza, ChickenPizza

Contains a method to set the topping to cheese, mushroom and chicken, respectively.

Driver

selectPizzaCategory(int pizzaCategory) - Create a suitable object and the relevant topping with respect to the selected pizza category (Chicken, Cheese, Veggie).

b) Create a menu as given in the image below:

```
*****Welcome to RPK Pizza*****
Enter your name:
John
*****WELCOME John*****
Select the Pizza size:
1)Large
2)Medium
3)Small
Large
Select the Pizza category:
1)Veggie Pizza
2) Cheese Pizza
3) Chicken Pizza
2
Select Extra ingredient 1:
1)Cheese
2)Sausages
3)Prawns
4)Mushroom
5)CuttleFish
6)Chicken
Sausages
Select Extra ingredient 2:
1)Cheese
2)Sausages
3)Prawns
4)Mushroom
5)CuttleFish
6)Chicken
Prawns
```

c) Create 6 Ingredient objects as follows:

Ingredient Name	Ingredient Price
Cheese	100.0
Sausage	60.0
Mushroom	40.0
Prawns	30.0
Cuttle Fish	80.0
Chicken	50.0

(**Hint- assign the ingredients as an array)

[P.T.O]

- d) Increment the price with respect to the pizza size as given below inside the previously implemented **calculateTotalPrice()** method.

Pizza Size	Incremented Amount
Large	800.0
Boston	500.0
New York	350.0

- e) Finally call the **calculateTotalPrice()**, **prepare()**, **pack()** and the **displayOrderDetails()** methods to get the final output shown below.

```
*****  
  
ORDER DETAILS  
  
Name: John  
Preparing your order..!!  
Selected Topping: Cheese,Sausages,Prawns  
Pizza size: Large  
Total: 1000.0  
Your Order is Ready..!!
```

[60 Marks]

_____END_____